

HEARNE, TX, USA

WMO: 720303

Lat: 30.872N

Lon: 96.622W

Elev: 87

StdP: 100.29

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53973

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	-1.6	-10.7	1.5	3.8	-8.0	1.9	4.4	11.3	10.5	10.4	10.0	2.3	330	0.399

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.2	37.7	23.5	36.8	23.4	35.6	23.5	27.0	32.0	26.3	31.3	25.7	30.8	3.3	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.5	28.7	25.1	20.5	27.9	24.5	19.7	27.6	85.2	32.2	82.0	31.3	79.2	30.6	36.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.1	7.2	DB	-6.2	40.8	2.4	3.2	-8.0	43.1	-9.4	45.0	-10.7	46.8	-12.5	49.1	(4)
			WB	-7.6	28.1	2.3	3.4	-9.3	30.6	-10.6	32.6	-11.9	34.5	-13.6	37.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.1	10.0	12.4	16.2	19.7	23.8	27.7	29.0	29.2	26.3	20.7	15.4	10.9	(6)
		DBStd	8.05	5.54	5.78	5.29	4.16	3.34	1.90	2.03	2.10	3.00	4.49	5.48	5.67	(7)
		HDD10.0	192	68	37	12	1	0	0	0	0	1	15	58		(8)
		HDD18.3	972	265	177	104	33	4	0	0	1	29	118	241		(9)
		CDD10.0	3893	68	105	204	291	428	531	587	594	489	333	177	86	(10)
		CDD18.3	1631	6	12	37	73	174	281	329	336	240	103	31	10	(11)
		CDH23.3	17499	37	69	243	606	1611	3132	3913	4109	2530	977	221	50	(12)
(13)	Wind	CDH26.7	8476	3	7	43	147	622	1559	2124	2306	1262	348	46	7	(13)
		WSAvg	2.7	2.9	3.4	3.4	3.4	3.0	2.6	2.3	2.1	1.9	2.2	2.5	2.6	(14)
		PrecAvg	973	78	76	78	81	93	88	54	60	83	110	85	87	(15)
		PrecMax	1434	269	212	195	245	202	322	204	145	205	292	241	285	(16)
		PrecMin	589	4	4	27	22	1	0	0	3	14	1	5	17	(17)
		PrecStd	229	56	54	39	50	55	71	46	47	53	83	64	65	(18)
		DB	26.6	27.6	30.8	32.3	35.0	37.7	42.5	39.9	37.7	33.8	30.1	27.5		(19)
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	16.6	17.9	21.7	20.9	22.4	22.3	29.0	23.6	21.8	22.0	22.6	17.4		(20)
		DB	24.0	25.7	27.6	30.1	32.9	36.1	37.5	37.8	36.1	32.1	27.8	24.8		(21)
		MCWB	16.3	17.7	19.5	20.7	22.7	22.4	23.9	23.7	22.7	22.2	20.6	18.2		(22)
		DB	22.1	23.2	26.1	28.0	31.8	34.8	36.3	37.0	34.5	30.2	26.0	22.5		(23)
		MCWB	16.3	17.5	18.8	20.5	22.8	23.2	23.6	23.8	23.0	21.5	19.0	17.5		(24)
		DB	19.7	21.6	24.3	26.9	30.4	33.1	35.0	35.8	32.9	28.8	23.8	20.9		(25)
		MCWB	15.2	17.2	18.2	19.8	22.4	23.9	24.1	23.9	22.7	20.4	18.8	17.2		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	20.1	21.2	23.3	24.4	26.3	27.6	30.9	27.2	26.6	25.2	26.5	21.6		(27)
		MCDB	22.4	23.8	27.2	28.4	30.4	32.2	38.7	33.4	30.9	29.1	26.9	23.4		(28)
		WB	19.0	20.2	21.7	23.3	25.2	26.5	27.3	26.5	25.7	24.1	22.5	20.3		(29)
		MCDB	21.9	23.0	25.0	26.8	29.0	30.8	32.7	32.6	30.4	28.6	25.2	22.6		(30)
		WB	17.8	19.1	20.6	22.4	24.4	25.8	26.3	25.7	25.2	23.4	21.2	19.0		(31)
		MCDB	20.7	21.8	23.9	26.1	28.5	30.2	31.3	31.6	30.0	27.3	23.9	21.5		(32)
		WB	15.9	18.0	19.7	21.5	23.6	25.2	25.5	25.2	24.6	22.6	20.0	17.6		(33)
(34)	Mean Daily Temperature Range	MCDB	18.9	20.6	22.8	25.4	27.8	29.7	30.9	30.9	29.5	26.3	23.1	20.2		(34)
		MDBR	12.4	11.8	12.3	12.2	11.1	11.0	11.7	12.2	12.1	13.3	12.6	12.0		(35)
		5% DB	15.1	13.2	12.6	12.7	12.2	13.2	13.7	13.6	13.9	13.5	13.4	13.5		(36)
		MCWBR	8.6	7.6	6.4	5.8	4.3	3.3	3.2	2.8	3.9	5.4	7.2	8.8		(37)
		5% WB	11.0	10.6	10.3	9.9	9.7	9.6	11.5	11.7	10.6	10.9	10.8	11.2		(38)
		MCWBR	8.4	7.4	6.8	5.7	4.5	3.9	3.9	3.0	3.4	5.6	7.3	8.8		(39)
		taub	0.323	0.342	0.365	0.412	0.454	0.445	0.461	0.450	0.417	0.374	0.364	0.326		(40)
(41)	Clear-Sky Solar Irradiance	taud	2.504	2.462	2.400	2.285	2.214	2.288	2.275	2.298	2.342	2.444	2.413	2.504		(41)
		Ebn at Noon	908	920	918	882	843	846	831	837	854	873	850	885		(42)
		Edh at Noon	88	101	115	134	145	134	135	130	120	102	96	84		(43)
		RadAvg	2.90	3.29	4.41	5.38	5.78	6.45	6.55	6.16	5.17	4.23	3.13	2.55		(44)
(45)	All-Sky Solar Radiation	RadStd	0.29	0.61	0.40	0.41	0.45	0.49	0.47	0.35	0.45	0.57	0.36	0.32		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86

Nomenclature: See separate page